When Diverse Paraphrases Do Not Help: LLM Data Augmentation for Portuguese Aspect-Based Sentiment Analysis

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PTT5, human data only · extraction (F1)
0.832
vs 0.637 for the best ABSAPT 2024 system, BERTimbau Large trained on GPT-augmented data

PTT5, human data only · sentiment (BAcc)
0.775
vs 0.652 for the best ABSAPT 2024 system, BERTimbau Large trained on GPT-augmented data

Augmented setups that beat the baseline
0 / 4
2 LLMs × 2 sub-tasks, 5 seeds each

1 Motivation

ABSA extracts each **aspect term** and its **polarity**. Portuguese data is scarce, so LLM paraphrases are an appealing way to scale it.

POSITIVE NEGATIVE

O **café da manhã** era ótimo, mas o **quarto** estava sujo.

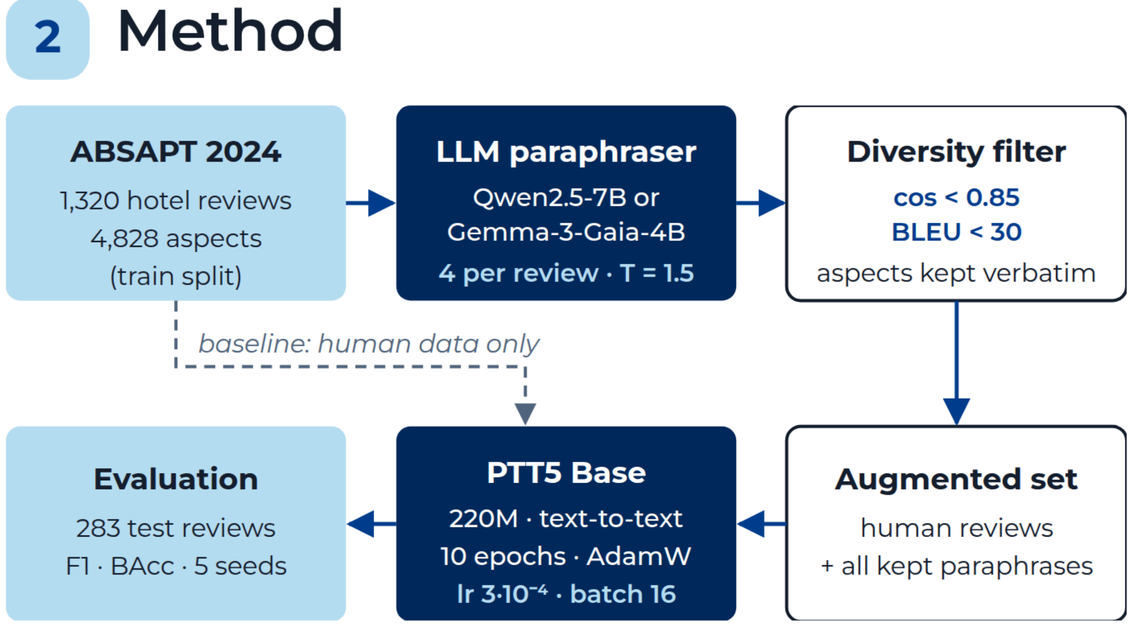
"The breakfast was great, but the room was dirty."

ATE · aspect term extraction | ASC · sentiment

RESEARCH QUESTION

Does in-context LLM paraphrasing yield data that improves a strong model on Portuguese ABSA?

2 Method



Tasks **ATE**: review → sorted list of aspect terms, micro-F1 (exact match). **ASC**: aspect + review → polarity, balanced accuracy.

Seeds 42 · 117 · 584 · 2026 · 9999. Each seed is one end-to-end run (generation → filter → training), so comparisons are paired.

Paraphraser	Kept / seed	Cosine	BLEU
Qwen2.5-7B	**1193 ± 8**	0.79 ± 0.07	11.2 ± 6.0
Gaia-4B	**1977 ± 30**	0.79 ± 0.07	9.3 ± 5.4

Table 1 — Kept of 5,280 candidates per seed (23% / 37%); similarity to the source.

3 Results

Augmentation **never helps**: the paired delta is negative in 4 of 5 seeds for every setup. Only the Qwen2.5-7B extraction drop is significant; sentiment stays within seed noise.

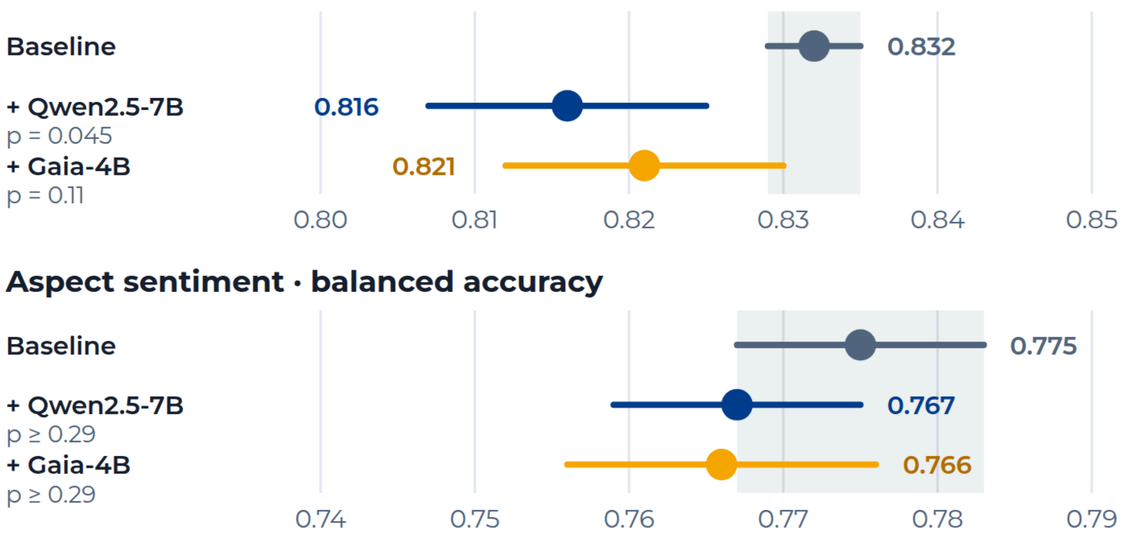


Figure 1 — Mean ± sd over five seeds; shaded band = baseline ± sd. Axes zoomed to a 0.05 range. p: paired t-test over seeds vs baseline.

4 Why no gain?

Human data is already the **closest to the test set**. The injected diversity is **mostly lexical**: no evidence of new semantic coverage, only surface variety that dilutes the training signal.

Training corpus	N	Cosine	BLEU
Human (original)	1317	**0.752**	**5.81**
Qwen2.5-7B	1193	0.725 ± 0.001	4.43 ± 0.08
Gaia-4B	1977	0.730 ± 0.001	4.30 ± 0.03

Table 2 — Proximity to the test set: max cosine (semantic) and BLEU to the 10 nearest test neighbours (surface). Train and test reviews are disjoint, so compare rows, not absolute BLEU. N = unique texts (3 near-duplicate human reviews merge: 1,320 → 1,317).

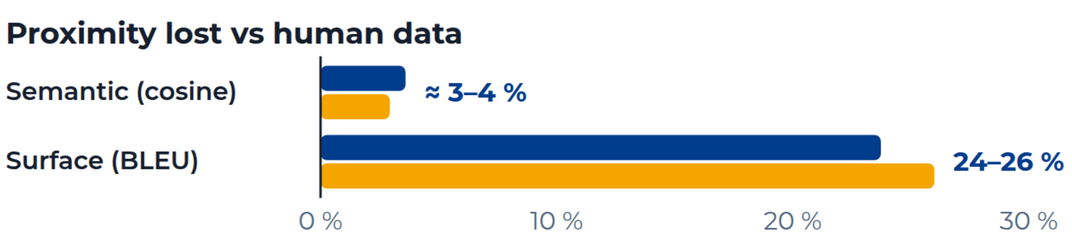


Figure 2 — Relative drop vs the human corpus (blue: Qwen2.5-7B, orange: Gaia-4B).

TAKE-HOME MESSAGE

Diverse paraphrases do not help: what matters is distance to the evaluation distribution, not distance to the seed.

And tuning anchoring, temperature and filters undercuts the appeal of automated LLM data scaling.

SCOPE & LIMITATIONS

- **No gain above ≈ 1 F1 point**, not a uniform drop: only the Qwen2.5-7B extraction drop is significant (5 seeds, fixed split).
- One hotel-review corpus, one target model, one augmentation policy (append all kept paraphrases).
- Hyperparameters fixed a priori, no dev set; two open-weight paraphrasers; cosine and BLEU are imperfect proxies.

Code & data: **github.com/AndrewwBC/INLG-2026** Contact: **abdcampos@inf.ufpel.edu.br**

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